

Movie Recommendation Engine

Internship Details

Internship: AI & Machine Learning Internship

Company: Synkoc IT Services Private Limited, Bengaluru

Intern: Suyanshi Patel

Duration: 6 Weeks

Abstract

With the rapid growth of streaming platforms, users have access to thousands of movies, making it difficult to discover content that matches their interests. The purpose of this project is to build a Movie Recommendation Engine that can intelligently suggest movies based on their similarities. The system uses machine learning and natural language processing techniques to analyze movie information and generate personalized recommendations.

Problem Statement

Users often spend a significant amount of time searching for movies to watch. Traditional search methods do not always provide personalized recommendations, leading to poor user experience and information overload. Therefore, there is a need for an intelligent recommendation system that can automatically suggest relevant movies to users.

Proposed Solution

A Content-Based Movie Recommendation System was developed using machine learning techniques. The system analyzes movie features such as genres, cast, directors, and keywords. It converts textual information into numerical representations using TF-IDF Vectorization and calculates similarities between movies using Cosine Similarity. Based on these similarity scores, the system recommends movies that closely match a user's interests.

Objectives

- To build an intelligent movie recommendation system.
 - To recommend movies based on content similarity.
 - To understand recommendation algorithms and NLP techniques.
 - To evaluate recommendation quality using Precision@10 and Recall@10.
 - To develop a production-ready recommendation workflow.
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Dataset Information

Dataset: TMDb 5000 Movies Dataset (Kaggle)

The dataset contains movie-related information including:

- Movie Title
- Genres
- Cast
- Crew
- Keywords
- Overview
- Release Information
- Ratings and Metadata

After merging the datasets, the final dataset contained approximately **4,809 movies and 23 features**.

Technologies Used

- Python
 - Pandas
 - NumPy
 - Scikit-learn
 - Jupyter Notebook
 - TF-IDF Vectorizer
 - Cosine Similarity
 - Matplotlib
 - Seaborn
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Project Development Phases

Phase 1: Movie Data Loading & EDA

Loaded the TMDb movie datasets and performed Exploratory Data Analysis (EDA) to understand the structure of the data, identify important features, and inspect missing values.

Phase 2: Content Feature Engineering

Combined important movie attributes such as genres, cast, directors, and keywords into a single feature representation called the content soup.

Phase 3: TF-IDF Similarity Matrix

Applied TF-IDF Vectorization to convert textual movie information into numerical feature vectors and created a similarity matrix representing relationships between movies.

Phase 4: Content-Based Recommender

Developed a recommendation engine that suggests movies similar to a selected movie based on content similarity scores.

Phase 5: Collaborative Filtering

Implemented collaborative filtering concepts to enhance recommendation quality and improve personalization.

Phase 6: Evaluation (Precision@10)

Evaluated the recommendation engine using Precision@10 and Recall@10 metrics to measure recommendation effectiveness and system performance.

Phase 7: Hybrid Recommendation System

Combined content-based and collaborative filtering approaches to build a more robust and accurate recommendation system.

Phase 8: Production Recommendation API

Designed a production-ready recommendation workflow capable of generating recommendations and providing explanations for recommended movies.

System Workflow

TMDB Dataset
↓
Data Loading & EDA
↓
Data Cleaning & Preprocessing
↓
Content Feature Engineering
↓
TF-IDF Vectorization
↓
Cosine Similarity Matrix
↓
Content-Based Recommender
↓
Collaborative Filtering
↓

Hybrid Recommendation System

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Evaluation (Precision@10)

↓

Production Recommendation API

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Personalized Movie Recommendations

Results

The recommendation engine successfully generated movie recommendations based on content similarity.

Performance Metrics:

- Average Precision@10: 0.806
- Average Recall@10: 0.032

The system demonstrated effective recommendation capabilities and provided relevant movie suggestions efficiently.

Conclusion

The Movie Recommendation Engine successfully addresses the problem of movie discovery by providing personalized recommendations based on movie content. The project strengthened understanding of machine learning workflows, natural language processing techniques, recommendation systems, and model evaluation strategies. The developed system can serve as a foundation for real-world recommendation platforms similar to Netflix and Amazon Prime Video.

Future Scope

- Develop a web-based user interface using Flask or Streamlit.
 - Integrate real user ratings and feedback.
 - Deploy the recommendation system on the cloud.
 - Improve recommendation quality using deep learning techniques.
 - Add multilingual recommendation capabilities.
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References

1. TMDB 5000 Movies Dataset – Kaggle
2. Scikit-learn Documentation

3. Pandas Documentation
4. NumPy Documentation
5. Research Papers on Recommendation Systems